MORPHISMS OF DECENT NOETHERIAN ALGEBRAIC SPACES FACTOR THROUGH THE RESIDUAL SPACE AT A POINT

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Residual space. definition-residual-space (same chapter, structural). The closed subspace t({x}) with its induced structure.
- Universally bounded fibres over the residual space. lemma-universally-bounded-finite-fibres (same chapter, carded L26058). The fibre of U -> X over the residual space is finite.

Source reference. Original lemma 06QZ: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3215–3215. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0H1Q: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1526–1526. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H1N: focused-seeds/geometry/S9-decent-factor-through-residual-space-noetherian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $x \in |X|$ be a point. Assume*

(1) *$|f|(|Y|)$ is contained in $\{x\} \subset |X|$,*
(2) *Y is reduced, and*
(3) *X is locally Noetherian.*

Then f factors through the residual space Z_x of X at x.

Proof. Preliminary remark: since $Z_x \to X$ is a monomorphism, it suffices to find a surjective étale morphism $Y' \to Y$ such that $Y' \to X$ factors through Z_x. A remark here is that Y' is reduced as well.

Let U be an affine scheme and let $U \to X$ be an étale morphism such that x is in the image of $|U| \to |X|$. Since X is locally Noetherian, U is a Noetherian affine scheme. By assumption (1) we see that $Y' = U \times_X Y \to Y$ is surjective

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as well as étale. Denote $E \subset |U|$ the set of points mapping to x. There are no nontrivial specializations between the elements of E, see Lemma 0H1Q. The morphism $Y' \to U$ maps $|Y'|$ into E. By our construction of Z_x in the proof of Lemma 06QZ we know that $\coprod_{u \in E} u \to X$ factors through Z_x. Hence it suffices to prove that $Y' \to U$ factors through $\coprod_{u \in E} u \to X$. After replacing Y' by an étale covering by a scheme (which we are allowed by our preliminary remark), this follows from Morphisms, Lemma 0H1N. $\square$